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ORIGINAL RESEARCH

Hydration Marker Diagnostic Accuracy to Identify Mild Intracellular and Extracellular Dehydration

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Identifying mild dehydration (≤2% of body mass) is important to prevent the negative effects of more severe dehydration on human health and performance. It is unknown whether a single hydration marker can identify both mild intracellular dehydration (ID) and extracellular dehydration (ED) with adequate diagnostic accuracy (≥0.7 receiver-operating characteristic–area under the curve [ROC-AUC]). Thus, in 15 young healthy men, the authors determined the diagnostic accuracy of 15 hydration markers after three randomized 48-hr trials; euhydration (water 36 $ml \cdot kg^{-1} \cdot day^{-1}$), ID caused by exercise and 48 hr of fluid restriction (water 2 $ml \cdot kg^{-1} \cdot day^{-1}$), and ED caused by a 4-hr diuretic-induced diuresis begun at 44 hr (Furosemide 0.65 mg/kg). Body mass was maintained on euhydration, and dehydration was mild on ID and ED (1.9% [0.5%] and 2.0% [0.3%] of body mass, respectively). Urine color, urine specific gravity, plasma osmolality, saliva flow rate, saliva osmolality, heart rate variability, and dry mouth identified ID (ROC-AUC; range 0.70–0.99), and postural heart rate change identified ED (ROC-AUC 0.82). Thirst 0–9 scale (ROC-AUC 0.97 and 0.78 for ID and ED) and urine osmolality (ROC-AUC 0.99 and 0.81 for ID and ED) identified both dehydration types. However, only the thirst 0–9 scale had a common dehydration threshold (≥4; sensitivity and specificity of 100%; 87% and 71%, 87% for ID and ED). In conclusion, using a common dehydration threshold ≥4, the thirst 0–9 scale identified mild intracellular and ED with adequate diagnostic accuracy. In young healthy adults', thirst 0–9 scale is a valid and practical dehydration screening tool.

***Keywords**:* hypohydration, plasma, ROC curve, saliva, tear, thirst, urine

No consensus currently exists on the best method to assess dehydration and prescribe fluid intake (Armstrong, 2007; Cheuvront & Kenefick, 2014; Cotter, Thornton, Lee, & Laursen, 2014). This is in part because dehydration is a complex condition that manifests as different types. When fluid intake is inadequate, and the concentration of body fluids lost is hypoosmotic relative to plasma (e.g., exercise sweat loss), the body fluid redistribution that occurs results in a relatively larger loss of intracellular than extracellular fluid (Sawka, 1992). Consequently, this type of dehydration is referred to as intracellular dehydration (ID) and is characterized by an increased plasma osmolality (hyperosmolality). In contrast, extracellular dehydration (ED) is caused by isoosmotic fluid loss and is characterized by volume depletion (hypovolemia) and the absence of hyperosmolality. ED often occurs when people are ill; take medications (e.g., diuretics); are immersed in water; or exposed to cold and/or hypoxia (Cheuvront & Kenefick, 2014; Cotter et al., 2014). Whether hydration markers identify ID or ED is likely to depend on the relationship between the marker and the distinct physiological characteristics of each dehydration type.

Potential candidate markers to identify both types of dehydration are urine, saliva, ratings of thirst and cardiovascular parameters, including resting and postural changes in heart rate and blood pressure, and heart rate variability (HRV; Cheuvront, Ely, Kenefick, Buller, Charkoudian, & Sawka, 2012; Cotter et al., 2014; Fitzsimons, 1976; Oliver, Laing, Wilson, Bilzon, & Walsh, 2008). These markers may respond directly to osmotic and volume stimuli or indirectly to the subsequent alterations in autonomic tone (Charkoudian, Eisenach, Joyner, Roberts, & Wick, 2005; Oliver et al., 2008; Sands & Layton, 2009). Although most of these hydration markers have shown promise to identify moderate and severe ID (>3% body mass; Armstrong et al., 1994, 2014; Cheuvront et al., 2012; Walsh, Laing, Oliver, Montague, Walters, & Bilzon, 2004), limited research has investigated the validity and diagnostic accuracy of these hydration markers to identify more mild ED or ID (≤2% of body mass). Mild dehydration is important to identify, as it is beyond this threshold that human performance has been consistently shown to decline (Cheuvront & Kenefick, 2014; Goulet, 2012; Savoie, Kenefick, Ely, Cheuvront, & Goulet, 2015).

The aim of this study was therefore to determine hydration marker diagnostic accuracy to identify mild ID and ED. Based on previous research examining hydration markers after moderate and

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severe dehydration (Cheuvront et al., 2012; Fortes et al., 2011; Oliver et al., 2008; Shirreffs, Merson, Fraser, & Archer, 2004), we hypothesized that urine, thirst, dry mouth, saliva, and HRV markers would identify both types of mild dehydration with adequate diagnostic accuracy (ROC-AUC ≥0.7; Hooper et al., 2016). Based on this research, we also hypothesized that plasma osmolality and tear osmolarity would identify mild ID, but not mild ED; and postural heart rate and blood pressure change would identify ED, but not ID.

Materials and Methods

Participants

Fifteen healthy men volunteered to complete this study (age 22.8 [5.4] years, height 180.4 [5.0] cm, mass 78.9 [8.6] kg, body mass index 24.2[1.8] kg/m^2; $\dot{V}O_2$max 52.3 [6.9] ml·kg^{-1}·min^{-1}). Participants were excluded if they were smokers; had abnormal blood chemistry or renal function; suffered from diabetes, asthma, bronchitis, epilepsy, hypertension, dental or oral disease; or were receiving any medication or treatment. Informed written consent was obtained from each participant. The study was approved by the Ethics committee of the School of Sport, Health and Exercise Sciences, Bangor University and adhered to the Declaration of Helsinki.

Preliminary Measures

As body mass loss during the 48-hr trials was the reference standard in this study, we standardized energy intake and physical activity 24 hr before and during trials. Energy intake was calculated as the product of resting metabolic rate and an estimated physical activity factor. Resting metabolic rate was estimated from anthropometry (Harris & Benedict, 1918) and adjusted by a general daily physical activity and diet-induced thermogenesis factor coefficient of 1.6, which was determined from the activities completed on trials (Todorovic & Micklewright, 2004). Participants were also habituated with the hydration assessment techniques and completed a graded cycle exercise test to determine their peak power output, which was used to prescribe the workload for the experimental trial cycling exercise (Excalibur Sport, Lode, The Netherlands).

Study Protocol

This study followed a crossover design. Separated by 7 days, participants completed three trials in a random order including a euhydrated control trial (EU), an ID trial, and an ED trial. Each trial consisted of a baseline hydration assessment, an exercise bout, one of the three 48-hr interventions, and a second hydration assessment (Figure 1). Hydration assessments and exercise were performed in an air-conditioned laboratory, temperature and humidity 19.4 (1.0) °C and 42% (6%), respectively.

The day before each experimental trial, participants abstained from alcohol, caffeine, or strenuous physical activity and consumed a standardized individually prescribed diet (energy and sodium intake 3,034 [245] kcal and 2.2 [0.1] g; 62%, 25%, and 13% carbohydrate, fats, and protein, respectively). Daily energy intake was the same for the duration of the trials except on Day 1 in which participants consumed additional food (391 [193] kcal) to replace energy expended during the cycling exercise. This was calculated from indirect calorimetry during the habituation visit cycling exercise test (Cortex MetaLyzer 3B, Leipzig, Germany).

On Day 1 of each trial, participants woke at 07:00 hr and drank water equal to 6 ml·kg^{-1}of body mass (471 [52] ml). On arrival to the laboratory at 08:00 hr, participants received a further bolus of water equal to 6 ml·kg^{-1}of body mass and a standardized breakfast (690 kcal, sodium 0.8 [0.1] g; 62%, 23%, and 15% carbohydrate, fats, and protein, respectively). To monitor and standardize physical activity on the trials, participants were fitted with pedometers and provided with step count targets (Digi-Walker SW200; Yamax, Tokyo, Japan). At 12:00 hr, participants returned to the laboratory

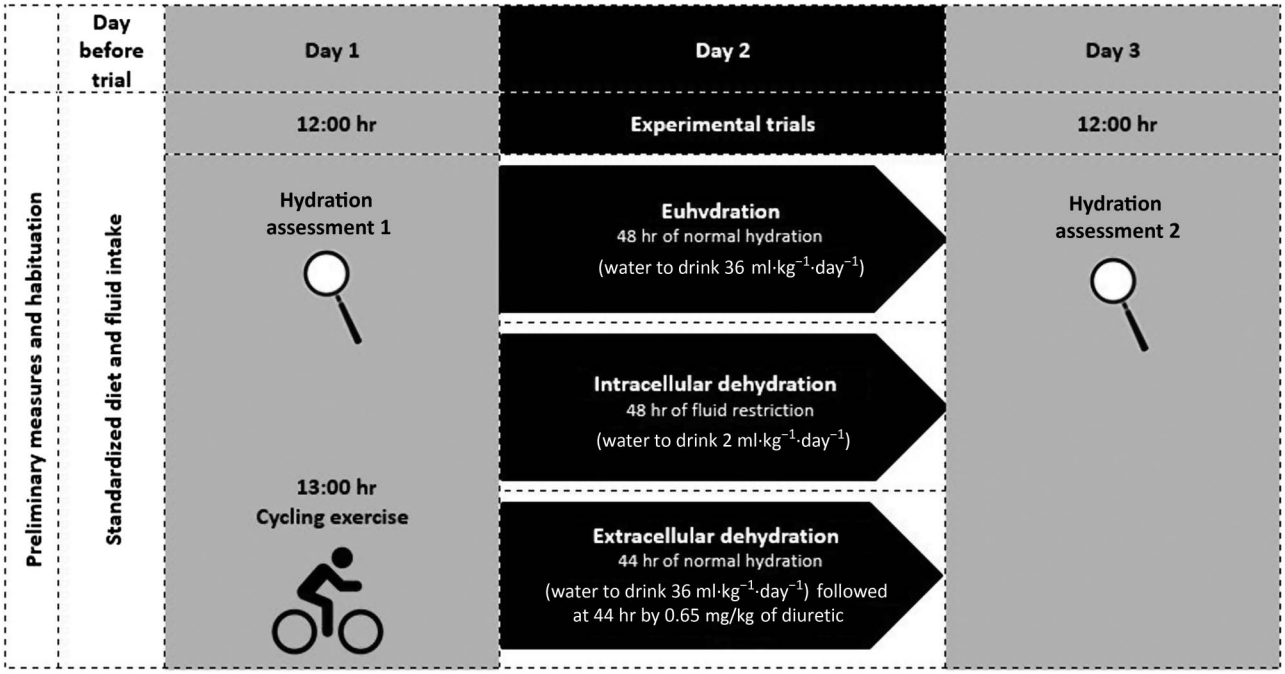

Figure 1 — Schematic representation of experimental trial. The cycling exercise intensity was 70% of peak power output until exhaustion. Hydration assessments and exercise were performed in an air-conditioned laboratory, temperature and humidity, 19.4 (1.0) °C and 42% (6%), respectively.

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for the baseline hydration assessment. Immediately after, dehydration was induced via cycling exercise at 70% peak power output until exhaustion. After the cycling exercise, the participants began one of three 48-hr trials. The calculated sweat loss from the cycling exercise was replaced with water on EU and ED but not on ID. Drinking water was restricted on ID to 2 ml/kg of body mass per day (total 314 [35] ml). In contrast, on EU and ED, participants drank water equal to 36 ml/kg of body mass per day (total for 48 hr 5,728 [600] ml). This fluid intake strategy was adapted from those previously used in our laboratory to maintain EU (Oliver, Laing, Wilson, Bilzon, & Walsh, 2007; Oliver et al., 2008; Walsh et al., 2004). On Day 3, participants reported to the laboratory at 07:30 hr. At 08:00 hr, and after a standardized breakfast, on ED participants consumed the diuretic Furosemide as a liquid equal to 0.65 mg/kg (51 [6] mg Frusol; Rosemont Pharmaceuticals, Leeds, United Kingdom). All urine voided between 08:00 and 12:00 hr was collected to measure total urine volume. At 12:00 hr on all trials, participants began the hydration assessment 2.

Hydration Assessments

Hydration markers were obtained in the same order on each trial and at each hydration assessment. First, participants completed subjective ratings of thirst and dry mouth on 100-mm visual analog scale, and the 0–9 thirst sensation scale (0 [*not thirsty at all*] to 9 [*very, very thirsty*]; Engell et al., 1987). Participants were instructed to respond to the scale based on how they felt at that moment. Second, a urine sample was collected in a container and immediately analyzed for urine color by an 8-point chart (Armstrong et al., 1994), urine specific gravity was measured in duplicate using a handheld refractometer (Atago, Tokyo, Japan), and urine osmolality was measured in triplicate by a freezing point depression osmometer (model 3300; Advanced Instruments, Norwood, MA). Third, nude body mass was determined to the nearest 50 g using a digital platform scale (model 705; Seca, Hamburg, Germany). Fourth, participants were fitted with a heart rate monitor (Polar RS800; Polar Electro, Kempele, Finland), after 2 min of seated rest, beat-to-beat heart rate was recorded for 10 min for the determination of HRV (Marek, 1996). All R–R series were extracted with a processing program (Polar Precision Performance; Polar Electro, Kempele, Finland) and analyzed in the time and frequency domain after automatic removal of occasional ectopic beats (Version 2.1; Kubios, BSAMIG, Kuopio, Finland). Fifth, the participants sat quietly for 5 min before a tear fluid sample was analyzed for tear osmolarity from the right eye as previously described (Fortes et al., 2011; TearLab™ Osmolarity System, San Diego, CA). Sixth, after 5 min supine rest, blood pressure and heart rate were recorded (Tango; SunTech Medical Ltd., Morrisville, NC). These measures were then repeated after exactly 1 min of standing for the determination of postural change measures of blood pressure and heart rate calculated as the difference between lying and standing measures. Seventh, a seated 5-min unstimulated saliva sample was collected for the determination of saliva flow rate and osmolality as previously described (Oliver et al., 2008). Finally, after 10 min seated rest, a venous blood sample was collected by venipuncture without venestasis into a vacutainer tube containing lithium heparin (Becton Dickinson, Franklin Lakes, NJ). This blood was immediately used to determine, in triplicate, hematocrit (packed cell volume) by microcentrifugation (Hawksley and Sons Ltd, Sussex, United Kingdom) and hemoglobin by automated analyzer (B-Hemoglobin, Hemocue, Sweden). Plasma volume change was then estimated from the change in hemoglobin and hematocrit values between hydration assessment 1 and assessment 2 (Dill & Costill, 1974; Strauss, Davis, Rosenbaum, & Rossmeil, 1951). The remaining blood was centrifuged at 1,500*g* for 10 min at 5 °C, and plasma was analyzed for osmolality in triplicate. If any of the intrasample osmolalities differed by more than 1%, a further sample was measured, and the mean of the four samples was used.

Statistical Analysis

Hydration marker diagnostic accuracy to identify mild ID and ED was determined from hydration assessment 2 data by ROC-AUC with 95% confidence intervals (MedCalc Software bvba, Ostend, Belgium) as recommended (Zweig & Campbell, 1993). Body mass change was used as the mild dehydration reference standard as it is a precise measure of body fluid change in controlled laboratory studies (Cheuvront, Ely, Kenefick, & Sawka, 2010; Oliver et al., 2008). Body mass loss was calculated on all trials to ensure euhydration was maintained on EU, and mild dehydration was achieved on ID and ED. A 1% threshold was used as this has previously been reported as the typical day-to-day variability of body mass in active men (Cheuvront et al., 2010). Hydration markers were also given a qualitative ROC-AUC descriptor that relates to the quantitative diagnostic accuracy statistic as poor (0.6), adequate (0.7), moderate (0.8), high (0.9), near perfect (0.95), and perfect (1.0; Obuchowski, Lieber, & Wians, 2004). For hydration markers to be considered to have adequate diagnostic accuracy, it has also previously been specified that ROC-AUC should be ≥0.7 (Hooper et al., 2016). A value of 0.5 indicates that a hydration marker has no better ability than chance to discriminate between EU and dehydration, whereas 1.0 indicates that the marker has perfect discrimination (Zweig & Campbell, 1993). A sample size of 15 was selected, to allow for dropout, and based on a balanced design (i.e., equal numbers of participants with and without dehydration) that indicated a sample size of 14 was sufficient to enable a marker with a diagnostic accuracy of ≥0.7 to be statistically discriminated from 0.5, that is, no better than chance. For hydration markers with adequate diagnostic accuracy (≥0.7), a secondary analysis was performed, where the Youden index was used to generate an objective mild dehydration threshold (Schisterman, Perkins, Liu, & Bondell, 2005). Hydration markers at the hydration assessments were also compared between trials by one-way analysis of variance with planned multiple comparisons by Tukey (version 6.0; GraphPad Prism, San Diego, CA). Unless stated, all values are represented as mean (*SD*), and statistical significance was accepted at $p < .05$.

Results

Hydration Assessment 1 and Trial Physical Activity

Standardization of pretrial fluid and energy intake was successful as indicated by consistent euhydrated status at hydration assessment 1 (CON, ID, and ED: plasma osmolality 287 [4], 289 [5], 287 [3] mOsm/kg, $p = .10$; urine specific gravity: 1.009 [0.004], 1.009 [0.004], 1.007 [0.003] $g \cdot ml^{-1}$, $p = .34$; body mass: 78.4 [8.4], 78.3 [8.3], 78.4 [8.7] kg, $p = .89$; coefficient of variation for plasma osmolality, urine specific gravity, and body mass were 1.0%, 0.3%, and 0.6%, respectively). Also similar on all trials was the cycling exercise time and sweat loss (CON, ID, and ED: time to exhaustion 1,200 [377], 1,339 [415], 1,323 [431] s, $p = .15$; sweat loss: 470 [200], 540 [150], 590 [200] ml, $p = .10$) and trial physical activity (CON, ID, and ED: 15,299 [4,172], 17,182 [5,106], 17,982 [4,625] $steps \cdot trial^{-1}$, $p = .08$).

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Hydration Assessment 2

Body mass, plasma osmolality, and plasma volume were stable on EU and support the decreased body mass on ID and ED represented mild dehydration and not an energy deficit ($p < .001$; Table 1). ID was confirmed by increased plasma osmolality (Table 1). ED was confirmed by decreased plasma volume without a change in plasma osmolality (Table 1). Furthermore, after the diuretic on ED, urine production was increased compared with EU and ID as expected (1,677 [338] vs. 772 [311] and 138 [54] ml, $p < .001$). Increased urine production on ED ceased before hydration assessment 2 as indicated by a similar urine volume on all trials at hydration assessment 2 (EU, ID, and ED: 143 [110], 97 [57], 189 [120] ml, $p = .13$). Compared with EU, the HRV index low- to high-frequency ratio was increased after ID but not ED (Table 1). Further cardiovascular and renal differences between ID and ED and the descriptive statistics for other hydration markers studied for diagnostic accuracy are outlined in Table 2.

Hydration Marker Diagnostic Accuracy

Thirst 0–9 and urine osmolality had adequate diagnostic accuracy to identify both mild ID and ED (Table 3). The diagnostic accuracy of these markers was near perfect to identify mild ID and moderate for mild ED. For thirst 0–9, the Youden index derived the same threshold for both mild ID and ED (≥4). The sensitivity and specificity of this threshold was 100% and 87% for ID and 71% and 87% for ED (Table 4). For urine osmolality, the Youden index derived two different thresholds depending on the type of dehydration (Table 4).

Several other hydration markers identified mild ID with adequate diagnostic accuracy (ROC-AUC ≥0.7; Table 3). The discriminatory accuracy was perfect for urine markers (color and specific gravity); near perfect for plasma osmolality; high for thirst (visual analog scale) and dry mouth (visual analog scale); and adequate for HRV, saliva flow rate, and osmolality. The mild ID thresholds for these hydration markers and their sensitivity and specificity to identify mild ID are shown in Table 4. In addition to thirst 0–9 scale and urine osmolality, postural change in heart rate was the only other hydration marker to identify mild ED with adequate diagnostic accuracy (ROC-AUC ≥0.7).

Discussion

This study extends current hydration marker understanding by using diagnostic accuracy statistics to evaluate several markers'

Table 1 Characterization of Experimental Hydration Status After Mild ID and ED

	EU	ID	ED
Body mass change (%)	0.0 (0.6)	−1.9 (0.5)**	−2.0 (0.3)**
Body mass change range (%)	+0.9 to −0.7	−1.2 to −2.9	−1.5 to −2.5
Body mass change (kg)	0.0 (0.5)	−1.5 (0.5)**	−1.6 (0.3)**
Blood volume change (%)	0.8 (4.7)	0.0 (4.3)	−3.5 (2.8)‡
Plasma volume change (%)	1.7 (6.2)	−0.3 (5.7)	−6.6 (4.0)‡‡
Plasma osmolality (mOsm/kg)	287 (4)	297 (7)††	286 (5)
HRV (LF/HF ratio)	1.8 (1.1)	3.4 (2.2)*	2.9 (2.1)

Note. ED = extracellular dehydration; EU = euhydration; HRV = heart rate variability; ID = intracellular dehydration; LF/HF ratio = low-/high-frequency HRV power ratio. Values are represented by mean (SD). Post hoc test differences indicated by *$p < .05$ versus EU, **$p < .01$ versus EU, ††$p < .01$ versus EU and ED, ‡$p < .01$ versus EU and ID, ‡‡$p < .01$ versus EU and ID.

Table 2 Hydration Markers After Mild ID and ED

	EU	ID	ED
Thirst (0–9)	3 (1)	6 (1)††	4 (1)*
Thirst (VAS)	33 (19)	69 (17)†	43 (17)
Dry mouth (VAS)	27 (17)	60 (21)††	36 (12)
Urine osmolality (mOsm/kg)	267 (138)	1,054 (127)††	402 (110)‡
Urine specific gravity (g/ml)	1.008 (0.004)	1.028 (0.005)††	1.010 (0.004)
Urine color (1–8)	2 (1)	6 (1)††	2 (1)
Saliva flow rate (µl/min)	365 (241)	196 (165)†	425 (321)
Saliva osmolality (mOsm/kg)	56 (12)	64 (13)†	55 (12)
Tear osmolality (mOsm/L)	296 (12)	300 (11)	292 (12)
Postural change in HR (b/min)	14 (8)	19 (10)	26 (12)‡
Postural change in SBP (mmHg)	8 (12)	4 (14)	0 (9)
Supine HR (b/min)	56 (10)	56 (12)	57 (15)
Supine SBP (mmHg)	112 (8)	111 (10)	108 (10)

Note. ED = extracellular dehydration; EU = euhydration; HR = heart rate; ID = intracellular dehydration; SBP, systolic blood pressure; VAS, visual analog scale. Values are represented by mean (*SD*). Post hoc test differences indicated by *$p < .01$ versus EU, †$p < .05$ versus EU and ED, ††$p < .01$ versus EU and ED, ‡$p < .05$ versus EU and ID.

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Table 3 Diagnostic Accuracy of Hydration Markers to Identify Mild ID and ED

	ID			ED		
Hydration marker	ROC-AUC	95% CI	SE	ROC-AUC	95% CI	SE
Urine osmolality (mOsm/kg)	0.99*	[0.88–0.99]	0.01	0.81*	[0.63–0.93]	0.09
Thirst (0–9)	0.97*	[0.84–0.99]	0.02	0.78*	[0.59–0.90]	0.08
Urine specific gravity (g/ml)	0.99*	[0.88–0.99]	0.01	0.68	[0.48–0.83]	0.10
Thirst (VAS)	0.92*	[0.76–0.98]	0.04	0.66	[0.47–0.83]	0.10
Dry mouth (VAS)	0.88*	[0.69–0.97]	0.06	0.66	[0.47–0.83]	0.10
Urine color (1–8)	0.99*	[0.88–0.99]	0.01	0.52	[0.33–0.70]	0.11
Plasma osmolality (mOsm/kg)	0.96*	[0.82–0.99]	0.03	0.53	[0.34–0.71]	0.11
Postural change in HR (b/min)	0.66	[0.47–0.82]	0.10	0.82*	[0.64–0.93]	0.08
HRV (LF/HF ratio)	0.72*	[0.52–0.87]	0.09	0.64	[0.45–0.81]	0.11
Saliva osmolality (mOsm/kg)	0.70*	[0.51–0.85]	0.09	0.55	[0.36–0.73]	0.11
Saliva flow rate (µl/min)	0.70*	[0.51–0.85]	0.09	0.55	[0.36–0.73]	0.11
Tear osmolality (mOsm/L)	0.61	[0.41–0.78]	0.11	0.61	[0.42–0.82]	0.11
Postural change in SBP (mmHg)	0.56	[0.37–0.74]	0.11	0.65	[0.46–0.82]	0.10
Supine SBP (mmHg)	0.56	[0.37–0.74]	0.11	0.64	[0.44–0.80]	0.11
Supine HR (b/min)	0.53	[0.34–0.72]	0.11	0.52	[0.33–0.70]	0.11

Note. CI = binomial exact confidence interval for AUC; ED = extracellular dehydration; HR = heart rate; HRV = HR variability; ID = intracellular dehydration; LF/HF ratio = low-/high-frequency HRV power ratio; ROC-AUC = receiver-operating characteristic-area under the curve; SBP = systolic blood pressure; SE = standard error (Hanley & McNeil, 1983); VAS = visual analog scale.
*The hydration biomarker identifies dehydration type better than chance. Hydration markers are ranked by combined diagnostic accuracy.

Table 4 Sensitivity and Specificity of Youden Derived Mild Dehydration Thresholds for Hydration Markers

	ID			ED		
Hydration marker	Mild dehydration threshold[a]	Sensitivity (%)	Specificity (%)	Mild dehydration threshold[a]	Sensitivity (%)	Specificity (%)
Urine osmolality (mOsm/kg)	>595	99	99	>341	80	87
Thirst (0–9)	≥4	99	87	≥4	71	87
Urine specific gravity (g/ml)	>1.016	99	99	No	–	–
Thirst (VAS)	>47	93	80	No	–	–
Dry mouth (VAS)	>40	79	80	No	–	–
Urine color (1–8)	≥4	99	99	No	–	–
Plasma osmolality (mOsm/kg)	≥291	93	87	No	–	–
Postural change in HR (b/min)	No	–	–	>14	93	60
Saliva osmolality (mOsm/kg)	≥57	73	67	No	–	–
Saliva flow rate (µl/min)	≤137	67	67	No	–	–
HRV (LF/HF ratio)	>2.8	57	93	No	–	–
Tear osmolality (mOsm/L)	No	–	–	No	–	–
Postural change in SBP (mmHg)	No	–	–	No	–	–
Supine HR (b/min)	No	–	–	No	–	–
Supine SBP (mmHg)	No	–	–	No	–	–

Note. ED = extracellular dehydration; HR= heart rate; HRV = HR variability; ID = intracellular dehydration; LF/HF ratio = low-/high-frequency HRV power ratio; ROC-AUC = receiver-operating characteristic-area under the curve; SBP = systolic blood pressure; VAS = visual analog scale.
[a]Youden derived mild dehydration threshold, where ROC-AUC ≥0.70.

validity to identify mild ID and ED. A particular strength of this study is the standardization of energy intake and physical activity during the experimental trials, which alongside the maintenance of body mass within typical day-to-day variation (Cheuvront et al., 2010) on the euhydrated control trial, provides confidence that individual participant body mass losses on ID and ED represent mild fluid rather than energy deficits. The primary finding of this study is that thirst 0–9 and urine osmolality were the only hydration markers with adequate diagnostic accuracy to identify both mild ID and ED, caused by exercise and 48 hr of fluid restriction and a 4-hr diuretic-induced diuresis, respectively. However, thirst 0–9 was the only marker with a common dehydration threshold to identify mild ID and ED (≥4 for ID and ED; Table 4).

Notably, the present study is the first to determine the validity of thirst ratings using diagnostic accuracy statistics (Table 3). As hypothesized, thirst had adequate diagnostic accuracy to identify

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both types of mild dehydration, which may be expected as it is the major homeostatic effector mechanism for restoring EU. Furthermore, that thirst identified both ID and ED is in agreement with known physiological regulators, whereby thirst is sensitive to changes in both osmotic and volume stimuli (Fitzsimons, 1976). Osmolality is the principal thirst regulator (Cheuvront & Kenefick, 2014), and this may explain the better diagnostic accuracy of thirst to identify ID than ED in this study (Table 3). Indeed, plasma osmolality was increased by 3.5% after ID, which exceeds the reported 2% osmotic threshold of thirst (Table 1; Zerbe & Robertson, 1983). The blood volume reduction is the most likely stimuli for the increase in thirst after mild ED as other thirst regulators, including, plasma osmolality, dry mouth, and saliva flow rate were similar after the ED and EU control trials.

In agreement with our hypothesis, plasma osmolality, saliva flow rate and osmolality, dry mouth, urine markers, and HRV showed adequate diagnostic accuracy to identify mild ID, whereas postural change in heart rate showed adequate diagnostic accuracy to identify mild ED (Table 3). The diagnostic accuracy of these markers compares favorably with that previously reported after more severe dehydration (ROC-AUC range, 0.89–0.98; Armstrong et al., 2014; Bartok, Schoeller, Sullivan, Clark, & Landry, 2004; Cheuvront et al., 2010, 2012). Identifying milder dehydration with similar diagnostic accuracy is practically advantageous. Contrary to our hypothesis, tear osmolarity did not identify ID and saliva osmolality, HRV, and postural blood pressure change did not identify ED with adequate diagnostic accuracy. The reason for the poorer than anticipated diagnostic accuracy in these markers compared with previous studies (equivalent to ≥3% of body mass; Ely et al., 2014; Fortes et al., 2011; Oliver et al., 2008) may relate to the smaller fluid-deficit and osmotic, volume and autonomic nervous system alterations. In addition, our HRV results highlight that autonomic nervous system alterations, when compared with EU, may be greater after ID than ED of the same magnitude ($p = .04$ EU vs. ID; $p = .14$ EU vs. ED; Table 1). Given the postulated role of autonomic nervous system in saliva control (Oliver et al., 2008), this may explain why saliva parameters' diagnostic accuracy was adequate to identify ID but not ED.

As thirst 0–9 and urine osmolality were the only markers to identify mild ID and ED with adequate diagnostic accuracy, they might be considered the most suitable to identify persons that require simple oral rehydration to prevent the negative consequences of more severe dehydration to performance. Practically, thirst 0–9 has some additional advantages to urine osmolality. This includes a common threshold to identify mild dehydration regardless of the dehydration type. Furthermore, thirst can be assessed instantly and is easy to assess repeatedly, which could be particularly useful to help guide daily fluid intake, and rehydration from exercise, with persons aiming to achieve thirst ratings below or equal to 4. Urine osmolality in contrast has a lengthy collection and analysis process that requires the collection of a urine sample, which is not always possible, and specialist laboratory analysis. We therefore recommend that the thirst 0–9 scale is used as the initial screening tool to identify mild dehydration, and where determining the type of dehydration is important, plasma osmolality and postural change in heart rate are used to confirm if the dehydration is intracellular or extracellular, respectively.

Our hydration marker findings should be considered carefully within the context they were obtained, that is, dehydration methods used, environmental conditions and population studied. Urine volume at the second hydration assessment was similar and suggests overall fluid balance was stable at the time when hydration marker diagnostic accuracy was determined. However, the time to mild dehydration was much longer on ID than ED (48 hr ID and 4 hr ED), and consequently, fluid redistribution between body fluid compartments may have been more complete after ID than ED (Sawka, 1992). As ED is typically acute, for example, when people are ill, take medications (e.g., diuretics), are immersed in water, or exposed to cold and/or hypoxia, it is a practical strength of this study that we determined hydration marker diagnostic accuracy after acute rather than chronic ED. In contrast, ID may occur chronically, as in this study, or acutely, for example, sweating from passive heating and/or exercise. As these different dehydration methods may influence fluid regulation and redistribution (Sawka, 1992) and hydration marker diagnostic accuracy, future studies are warranted comparing the diagnostic accuracy of hydration markers to identify different dehydration methods, particularly that occur across different time courses. As in the present study, these future studies would benefit from measuring fluid compartments to confirm fluid redistribution by isotope or dye tracer techniques (e.g., bromide, Evans blue). Given the potential of thirst as a practical hydration marker, studies are needed to compare the diagnostic accuracy of thirst to identify acute and chronic mild ID. These studies are important as causes of acute ID including exercise, and exposure to hot and dry environments may alter thirst independently of dehydration due to direct effects of high ventilation, heat, and drying of the oral cavity. Future studies should also determine the diagnostic accuracy of thirst in other populations, for example, females, children, and the elderly. In the elderly, the diagnostic accuracy of thirst may be poorer than in young healthy adults as ageing and disease impair kidney and saliva gland function; in addition, the elderly are more likely to take medications that induce dry mouth which may alter thirst independently of dehydration (Kenney & Chiu, 2001; Scully, 2003). Furthermore, elderly persons with dementia and young children may not interpret the thirst scale as young healthy adults.

In conclusion, thirst 0–9 scale was the only hydration marker, with a common dehydration threshold, to identify both mild ID and ED with adequate diagnostic accuracy in young healthy men, residing in a thermoneutral environment. The practical utility of thirst is reinforced because it is a free and simple to use hydration marker that could also guide fluid intake to maintain EU.

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